

Van-Thinh To

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EDUCATION

Erasmus Mundus Scholar - Double Degree Master's Program

University Paris Cité (France)

2025 - expected 2027

Master of Bioinformatics

GPA: 18.05 / 20.00

University of Milan (Italy)

2025 - expected 2027

Master of Chemistry

GPA: 18.05 / 20.00

University of Medicine and Pharmacy at Ho Chi Minh City (Vietnam)

2019 - 2024

Bachelor of Pharmacy (5-year program)

GPA: 3.31 / 4.00

Ranking: Top 4%

Thesis: A Knowledge-Guided Graph Self-Supervised Learning for Molecular Property Predictions

Grade: 10 / 10

RESEARCH EXPERIENCE

University of Paris Cité (France)

Supervisor: Prof. Olivier Taboureaux and Assis. Prof. Viet-Khoa Tran-Nguyen

Master researcher

1. PubChem and ChEMBL Benchmarking Data Sets

Sep 2025 - present

- Automating large-scale retrieval of bioactivity data for all protein targets from PubChem and ChEMBL.
- Identifying dark chemical matter compounds for use as assumed inactives.
- Conducting virtual screening benchmarking studies across selected protein targets.

2. Target-specific machine learning scoring functions for cystathionine β -synthase inhibitors

Jan 2026

- Curated data set from PubChem and ChEMBL.

University of Medicine and Pharmacy at Ho Chi Minh City (Vietnam)

Supervisor: Assoc. Prof. Tuyen Ngoc Truong and PhD. Researcher Tieu-Long Phan

Undergraduate researcher

1. Generative AI for dual targets of EGFR and VEGFR

May 2024 - August 2025

- Developed a knowledge-guided graph-based scoring function to evaluate generated molecules.

2. A knowledge-guided graph for molecular representation

Dec 2023 - Jul 2024

- Introduced pre-text tasks for self-supervised learning guided by orbital information.
- Proposed a novel knowledge-guided graph (KGG) utilizing a hierarchical molecular graph, defragmentation, and two knowledge vectors that incorporate orbital information for molecular representation.
- A 2.4% improvement in ROC-AUC for classification tasks, along with a 39%, and 14% reduction in RMSE, and MAE respectively for regression tasks, compared to a state-of-the-art model on the MoleculeNet dataset.

3. A novel application of machine learning in virtual screening of HIV integrase inhibitors

Jun 2022 - Jan 2024

- Identified the applicability domain of the machine learning models using Principal Component Analysis for dimension reduction, coupled with convex hull techniques.

4. Application of molecular similarity, neural network and GNINA molecular docking in virtual screening of PD-L1 inhibitors

Jun 2023 - Dec 2023

- Proposed an artificial neural network model, and a molecular similarity assessment; utilized GNINA 1.0 molecular docking.
- Developed an automated pipeline for the molecular similarity model.
- Screened 15,235 DrugBank repurposing compounds and identified 22 hit compounds.

5. Machine learning and Deep neural networks with consensus molecular docking for enhanced potency prediction of ALK inhibitors

Dec 2022 - Jun 2023

- Trained an ensemble model including a graph neural network, an artificial neural network and a XGBoost model for the prediction of potential Anaplastic Lymphoma Kinase inhibitors.

TEACHING EXPERIENCE

University of Medicine and Pharmacy at Ho Chi Minh City (Vietnam)

Aug 2024 - Aug 2025

Contracted teaching assistant and research mentoring - Department of Organic Chemistry, Faculty of Pharmacy

- Wrote research proposals for four university-funded projects, each awarded 1200 USD.
- Prepared training materials on computer-aided drug design for undergraduate students, such as **RDkit**, fingerprints, molecular descriptors, ANN, GNN.
- Mentored one undergraduate student on thesis project focused on self-supervised learning.
- Taught practical organic chemistry, with a focus on laboratory equipment proficiency, ChemDraw usage, lab techniques and synthesis skills.

SKILLS

- **Cheminformatics:** **RDkit**; Machine Learning (**Scikit-learn**); Deep learning (**TensorFlow**, **Pytorch**); Molecular Docking (**GNINA**); Generative diffusion model (**IRDiff**); Property-matched decoy generation (**DeepCoy**); Databases (**PBD**, **PubChem**, **ChEMBL**); Visualization (**PyMOL**)
- **Programming skills:** Programming language (Python, R, C), Linux terminal, Pytest, Git, Package management, deployment
- **IELTS:** Overall 7.0 (Academic Version) Nov 2024

TRAINING

Vienna Summer School on Drug Design (Austria)

Sep 2023

- Gained foundational training in drug discovery using docking, pharmacophore and molecular simulation tools (LigandScout, KNIME, OpenEye)
- Learned concepts of techniques in drug discovery, such as generative models.

SELECTED PUBLICATIONS

Find all here

1. **To, V.-T.**, Van Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. F. & Truong, T. N. KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions. *Journal of Chemical Information and Modeling*. doi:10.1021/acs.jcim.5c01068 (2025)
2. Van Nguyen, P.-C., **To, V.-T.**, Tran, N.-V. N., Phan, T.-L., Truong, T. N., Gärtner, T., Merkle, D. & Stadler, P. F. SynCat: molecule-level attention graph neural network for precise reaction classification. *Digital Discovery*. doi:10.1039/D5DD00367A (2026)
3. **To, V.-T.**, Phan, T.-L., Doan, B.-V. N., Nguyen, P.-C. V., Le, Q.-H. N., Nguyen, H.-H., Trinh, T.-C. & Truong, T. N. Innovative virtual screening of PD-L1 inhibitors: the synergy of molecular similarity, neural networks and GNINA docking. *Future Medicinal Chemistry*, 1–12. doi:10.1080/17568919.2024.2389773 (2024)
4. Phan, T.-M., Phan, T.-L., Van-Nguyen, P.-C., Le, H. S. L., **To, V.-T.**, Truong, T. N., Merkle, D. & Stadler, P. F. ProQSAR: A Modular and Reproducible Framework for Small-Data QSAR Modeling with Fit-and-Use Models. *Journal of cheminformatics* (2026) (accepted)
5. Phan, T.-L., Trinh, T.-C., **To, V.-T.**, Pham, T.-A., Van Nguyen, P.-C., Phan, T.-M. & Truong, T. N. Novel machine learning approach toward classification model of HIV-1 integrase inhibitors. *RSC advances* **14**, 14506–14513. doi:10.1039/D4RA02231A (2024)
6. Truong, G.-B., Pham, T.-A., **To, V.-T.**, Le, H.-S. L., Nguyen, V. P.-C., Trinh, T.-C., Phan, T.-L. & Truong, T. N. Discovery of VEGFR-2 Inhibitors employing Junction Tree Variational Encoder with Local Latent Space Bayesian Optimization and Gradient Ascent Exploration. *ACS Omega*. doi:10.1021/acsomega.4c07689 (2024)
7. Trinh, T.-C., Phan, T.-L., **To, V.-T.**, Pham, T.-A., Truong, G.-B., Le, L. H. S., Tran, X.-T. D. & Truong, T. N. Synergy of advanced machine learning and deep neural networks with consensus molecular docking for virtual screening of anaplastic lymphoma kinase inhibitors. *Journal of Computer-Aided Molecular Design* **39**, 79. doi:10.1007/s10822-025-00657-6 (2025)

TALKS

1. **To, V.-T.**, Van-Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. & Truong, T. *KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions* in *41st Scientific and Technical Conference of Pharmacy* (2025)(Oral - Presenter)
2. **To, V.-T.**, Phan, T.-L. & Ngoc Doan, B.-V. *Application of Molecular Similarity and Artificial Neural Networks for PD-L1 inhibitors Virtual Screening* in *ECMC2023: 9th International Electronic Conference on Medicinal Chemistry* (2023) (Poster - Presenter)

SELECTED AWARDS

- **Pharmacy Scholarship for Elite Students** *2019 - 2024*
Awarded a \$5000 stipend over 5 years, covering full tuition (acceptance rate: 10%)
- **Second prize in the nationwide Vietnamese Chemistry Olympiad** *2018, 2019*
The biggest Chemistry competition for high school students in Vietnam with 400+ participants, ranking 42 overall
- **Odon Vallet Scholarship** *2019, 2024*
Awarded to outstanding high school, university, and graduate students in Vietnam, among only over 1,000 recipients nationwide.

REFERENCES

Assoc. Prof. Tuyen Ngoc Truong

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PhD. Researcher Tieu-Long Phan

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